APPLE INSPECTION STANDARDS

"Quality Control Service at Destination"
Fresh Apples

August 2022

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1. Glossary

- ✓ **Box Label Information:** Information present in every box which include specific information about the fruit such as variety, size, packing date, grower, category.
- ✓ **Commodity:** This term is used as a synonym for species. Ex: Apples, Cherries, Oranges, etc.
- ✓ **Condition defects:** All of the damage that has evolution over time, due to temperature, duration of the transportation, etc. Ex: Open wounds, putrefaction, damages, etc.
- ✓ **Container:** is a cargo recipient used in sea, terrestrial or multimodal transportation. For our interests, it is used to transport fruit and vegetables from the loading center to its destination.
- ✓ **Container number:** Each container has an individual number that identifies it. The numbers are composed of 4 letters, 6 numbers and an identification digit (ex: SUDU 123456-7). The four letters correspond to a specific navigation company. For example, "SUDU" corresponds to the company "Hamburg Sud".
- ✓ **Lamp:** Is the ventilation regulator that is found in the posterior part of the container which allows the gas exchange of the environmental atmosphere in the inside of the container.
- ✓ **Package/box:** What protects the fruit during the trip, it can be of paper, plastic or wood.
- ✓ **Packing list:** List corresponding to a specific container that indicated the pallets per producer, size and number of boxes.
- ✓ **Pallet:** Set of boxes packed and consolidated in a pallet base, labelled with a card and sheets.
- ✓ **Pallet Number:** Internal and self-copying register that identifies and individualizes the information of a pallet within a list.
- ✓ **Quality defects:** All of the damage that do not have an evolution over time, where the grade of the damage does not vary. For example: size, presence of foreign elements, healed wounds, etc.
- ✓ **Sample:** Selection of boxes within the container which should be inspected, whenever possible, that will cover all of the distinct parameters such as varieties, packing date, and producers, with the objective that the sample will be the most representative possible.
- ✓ **Thermograph:** Sensor that registers the temperature of the container from the origin to destination.

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2. Sampling

The objective of the sampling is to have a representative selection of samples over the lot to be inspected. To choose a sample must have in consideration the following parameters:

- Variety
- Category
- Grower
- Size
- Packing Date

Not always all these parameters will be available, in that case the sample selection has to be considering the ones that are available in the packing list, pallet number information or Box Label information.

For a regular apple inspection 6 boxes from 6 different pallets must be chosen, 1 box from 1 pallet. If the selected samples are not enough to make the inspection representative, more can be taken.

Sample selection restrictions:

- The chosen sample has to be taken from the 2nd row or below, never the first one.
- Must not perform inspection over a previously inspected box, it can determined by looking if the box was previously open or if it has a stamp over the box showing that it was inspected by other quality control entity.

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3. Inspection Instruments

Thermometer

The pulp temperature has to be taken, if possible, at the download of the container to have record of the temperature of arrival, in case the fruit is stored in a cold room also important to have record of the temperature the fruit has in that moment. The optimum travel temperature for apples is between 0°C to 1°C



Refractometer:

The refractometer is used to measure the brix (°Bx) of the fruit.

- ✓ Take one piece of normal looking fruit, squeeze some juice onto the prism, if the reading is unusual take a new piece to see if it was not a unique case.
- ✓ Make sure the prism is cleaned after each measurement; use water. Make sure the prism is dry before each measurement; use paper/cloth.





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Penetrometer

Instrument to measure the firmness of the fruit, the apple is expected to arrive at 14lb minimum at the moment of arrival at the warehouse.



Penetrometer using procedure:

- ✓ The peel or skin should be first removed on both sides before measuring the firmness with the peeler that comes in the penetrometer.
- ✓ Depth of cut should be uniform and not be too deep, the deeper the cut the higher the reading
- ✓ **Never test a bruised area of the fruit!**
- ✓ For testing, the fruit should be placed on a hard surface rather than being hand held;
- ✓ For accurate readings we **test the same fruit 2 times from the opposite sides of the ecuatorial part.**
- ✓ The plunger should be inserted in the fruit to the line inscribed on the plunger;
- ✓ **Uniformity of executing the firmness test is essential.**

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4. Group Clasification

Every group have different parameters that determine the classification, among them are the color covering, color intensity, cosmetics defects presence and varietal aspects.

Color Covering: Surface covered by color



Color Intensity: Color Strength



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Red Delicious

Varieties: Early Red One, Red Chief, Scarlet Spur, Super Chief, Stark Spur Ultrared



RED DELICIOUS		
SIZE	MIN WEIGHT (GR)	MAX WEIGHT (GR)
64	283	317
72	253	283
80	229	253
88	205	229
100	181	205
113	162	181
125	146	162
138	134	146
150	123	134
163	115	123
175	103	114
198	93	103
216	85	93

RED DELICIOUS GROUP		
CATEGORY	COLOR	DEFECTS
PREMIUM / SUPER	90-100%	WITHOUT
EXTRA FANCY	75-90%	WITHOUT
FANCY	>75%	WITHOUT
CHOICE	>65-75%	SLIGHT

Granny Smith

Variety: Granny Smith



GRANNY SMITH		
SIZE	MIN WEIGHT (GR)	MAX WEIGHT (GR)
64	283	317
72	253	283
80	229	253
88	205	229
100	181	205
113	162	181
125	146	162
138	134	146
150	123	134
163	115	123
175	103	114
198	93	103
216	85	93

GRANNY GROUP		
CATEGORY	APPEARANCE	DEFECTS
EXTRA FANCY	CLEAN	WITHOUT
FANCY	WITISH AREAS (VARIETAL), NO BLUSH	WITHOUT
EXTRA FANCY A	WITISH AREAS (VARIETAL), SOFT BLUSH	SLIGHT

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Galas

Varieties: Royal Gala, Galaxy, Brookfield Gala



ROYAL GALA		
SIZE	MIN WEIGHT (GR)	MAX WEIGHT (GR)
216	70	94
198	94	106
175	106	118
163	118	126
150	126	137
135	137	148
125	148	165
110	165	176
100	176	194
90	194	220
80	220	244
70	244	298
64	298	360
56	360	450

GALA GROUP		
CATEGORY	COVERING	INTENSITY
PREMIUM	75-100%	WITH
EXTRA FANCY	50 -100%	WITHOUT
CHOICE	50 - 75%	WITHOUT
FANCY	35 - 50%	WITH

Fuji Raku Raku

Varieties:Fuji Raku Raku, Fuji Raku Raku Grated, Fuji Raku Raku Coverd



FUJI RAKU RAKU		
SIZE	MIN WEIGHT (GR)	MAX WEIGHT (GR)
216	89	96
198	96	112
175	112	124
163	124	132
150	132	144
138	144	156
125	156	177
113	177	188
100	188	218
88	218	254
80	254	268
72	268	310
64	310	346
56	346	410
48	410	470
44	470	515
40	515	620

FUJI GROUP		
CATEGORY	COLOR	DEFECTS
PREMIUM	>70%	WITHOUT
EXTRA FANCY	50-70%	WITHOUT
EXTRA FANCY (CHOICE)	50-100%	SLIGHT
FANCY	30-50%	WITHOUT
FANCY (CHOICE)	30-50%	SLIGHT
CHOICE (CAT 1)	10-30%	WITHOUT

Pink Lady (Cripps Pink CV)

Variety: Pink Lady



PINK LADY		
SIZE	MIN WEIGHT (GR)	MAX WEIGHT (GR)
198	85	100
175	100	106
163	106	110
150	110	122
135	122	135
120	135	154
110	154	170
100	170	183
90	183	214
80	214	240
70	240	260
64	260	285
56	285	350
48	350	450

PINK LADY GROUP		
CATEGORY	COLOR	DEFECTS
PREMIUM	>40%	WITHOUT
EXTRA FANCY	30-40%	WITHOUT
EXTRA FANCY (CHOICE)	30-40%	SLIGHT
FANCY	20-40%	WITHOUT

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5. Tolerances

QC SCORE	Very Good	Good	Fair	Poor	Very Poor
	1	2	3	4	5
QUALITY DEFECTS	%	%	%	%	%
General Appearance	1	2	3	4	5
Undersize (Gr)	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Color Red Variety (A, B, C, D)					
Under Color	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Skin Defect	> 0 - 5	> 5 - 10	> 10 - 20	> 20 -30	> 30
Russet / Scars Slight / Moderate	> 0 - 5	> 5 - 15	> 15 - 25	> 25 - 35	> 35
Russet / Scars Severe	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Foreign Material / Dirty	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Mechanical Damage	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Peduncle Broken / Absence	> 0 - 5	> 5 - 15	> 15 - 30	> 30 - 45	> 45
Misshape	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Sun Burn	> 0 - 5	> 5 - 10	> 10 - 20	> 20 - 30	> 30
Total Quality defects	**> 0 - 5**	**> 5 - 10**	**> 10 - 20**	**> 20 - 30**	**> 30**
QUALITY CONDITION	%	%	%	%	%
Decay	0	> 0 - 2	> 2 - 4	> 4 - 10	> 10
Mold	0	> 0 - 2	> 2 - 4	> 4 - 10	> 10
Soft	> 0 - 1	> 1 - 4	> 4 - 8	> 8 - 15	> 15
Bruises	> 0 - 1	> 1 - 10	> 10 - 20	> 20 - 30	> 30
Bitter Pit	> 0 - 1	> 1 - 4	> 4 - 8	> 8 - 15	> 15
Scald	> 0 - 1	> 1 - 5	> 5 - 10	> 10 - 16	> 16
Dry Fruit / Shriveling	> 0 -1	> 1 - 5	> 5 - 10	> 10 - 16	> 16
Watercore Damage	> 0 - 1	> 1 - 5	> 5 - 10	> 10 - 16	> 16
Core Flush	> 0 - 1	> 1 - 4	> 4 - 8	> 8 - 15	> 15
Chilling / Freezing Damage	> 0 - 1	> 1 - 5	> 5 - 10	> 10 - 16	> 16
Internal Breakdown	> 0 - 1	> 1 - 10	> 10 - 20	> 20 - 30	> 30
Open Wound	> 0 - 1	> 1 - 5	> 5 - 10	> 10 - 16	> 16
Insect Presence / Damage	> 0 - 1	> 1 - 5	> 5 - 10	> 10 - 16	> 16
Total Condition defects	**> 0 - 1**	**> 1 - 10**	**> 10 - 20**	**> 20 - 30**	**> 30**

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6. Quality Defects: All of damage that does not have an evolution over time, where the grade of the damage does not vary. For example: size, presence of foreign elements, healed wounds, etc. All the parameters that don't have evolution over time, comes from the pre-harvest period and harvest handling.

Under Size (Gr): Size is determined by maximum diameter of the equatorial section or by weight of each apple. Every group has a sizing table for reference.

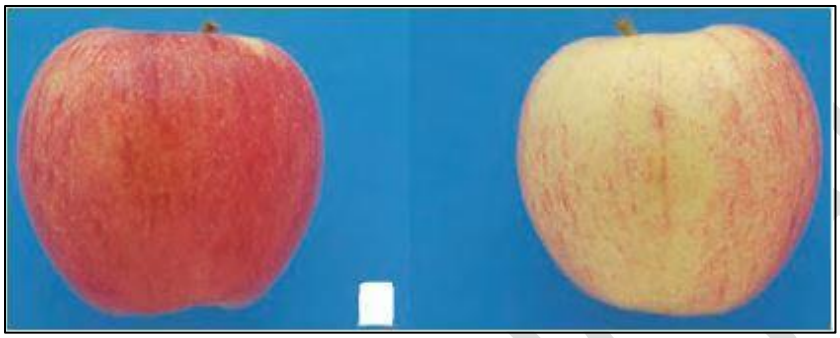


Color Red Variety: The color covering is described for every group.



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Under Color (Discoloration): When the color of a piece of fruit is not conform the color characteristics of the variety, it may be considered as a defect. However moderate discoloration due to growing conditions and normal ripening of the fruit, is not considered as a defect.



Skin Defect*:* For all defects founds in the apple skin not mentioned in this manual. Some superficial skin defects are allowed for Class I. The injuries do not exceed 1 cm2 in length, and 2 cm2 for the class II.







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Russet / Scars Slight / Moderate: The result of moderate damage to the epidermal cells of the apple. Some superficial russets, scars or scratches are allowed for Class I. The injuries should not exceed 1 cm2 in length.





Russet / Scars Severe*:* The result of severe damage to the epidermal cells of the apple. Some superficial russets, scars or scratches are allowed for Class I. The injuries do not exceed 2 cm2 in length. For scars due to scab, there is only a small. Russeting not exceeding 1/5th of the surface is allowed.







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Foreign Material / Dirty: Fruit with plant material or remains of residue. Also, secretion due to pests like aphids or mealybugs is considered as unclean.



Mechanical Damage: Damage appears as a result of all processes that the fruit be injured on the skin, in the crops, or in post-harvest. This is allowed, provided it does not exceed 1 cm^2 and does not affect the fruit flesh. Mostly noticed on one size of the fruit.

- ✓ Hint: *Hail Damage is a example of* Mechanical Damage. This damage appears as small dens on the peel of the fruit like shown below but increases in size when the fruit grows.



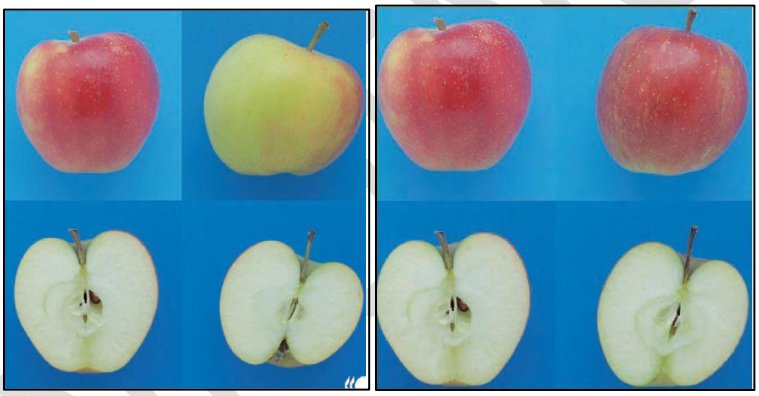


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Peduncle Broken / Absence*:* When the apple shows damage or absence of peduncle.

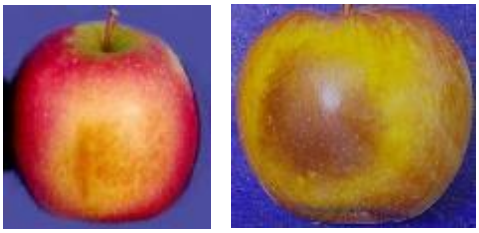


Misshape: Fruit with a distinct shape compared with the natural. Also, it is not symmetric. Slight malformation is allowed, provided the fruit keeps the characteristics belonging to the variety.



Sun Burn*:* It is often observed as a golden to bronze skin line discoloration caused by the exposure to sunlight on one side of the fruit. For Class I they are allowed, provided they do not exceed 20% of the surface of the fruit and will not affect the fruit flesh.





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7. **Condition Defects**: All of damage that has evolution over time, due to temperature, duration and conditions of transportation, etc. For example: Open wounds, putrefaction, pitting, etc. All the parameters suffer deterioration over time and reduce shelf life.

Decay: During the decay process, the fruit have a brown color, disintegration with an aqueous aspect and a fetid odor resulting from fermentation. Sometimes the fruit rot is present in the core /cavity of the fruit which can only be observed while cutting the fruit.



Mold: These defects are the presence of a fungus that grows when the fruit has injuries due to post-harvest handling or treatment.



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Soft*:* Fruit with low firmness.



Bruises: Mostly caused due a rough post-harvest handling of the fruit. The bruises will show a brown pulp under the skin.



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Bitterpit: This defect is also called Cork Spot and appears as dark sunken spots on the rind of the apple; the flesh just below the skin is brown and corky. The disorder starts internally and finally causes external blemishes.

The varieties Golden delicious and Granny Smith are in particular susceptible.

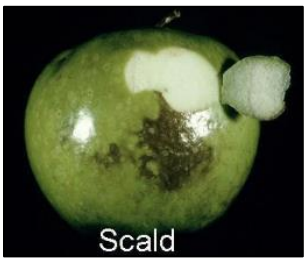




Scald: The affected fruit shows a browning of the skin, mostly seen on the shaded side (greener side) of the fruit. Often it only affects the skin and the fruit flesh remains intact. However, the severity of storage scald depends on the apple variety and type of scald.

- ✓ *Symptoms:* Usually the symptoms of ordinary scald are not evident on fruits during the first three months of storage at -1 - 0°C but may develop rapidly after that if the fruits are transferred to warm air.
- ✓ *Hint:* Symptoms become evident on the fruits after 4-5 months at cold-storage temperatures, and they intensify after the fruits are moved to warm air.





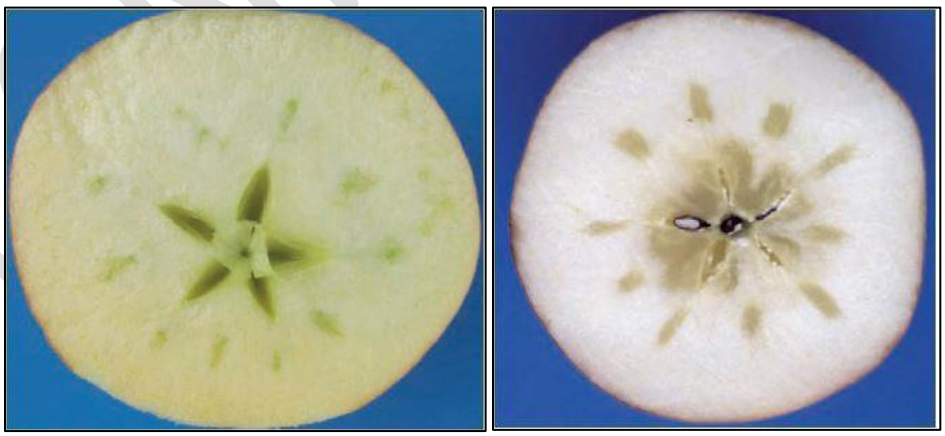
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Dry Fruit (Shriveling): Observed as wrinkling of the peel of the fruit due to dehydration or ageing fruit. It will result in degradation of the fruit in case the defect becomes visible.



Watercore Damage*:* This defect is observed as glassy, waterlogged sections in the fruit flesh of the apple. In mild cases, it is mostly observed around the core of the fruit. In severe cases, it affects the whole fruit and may be visible from the outside. Watercore often increases rapidly as apples become over mature.

- ✓ Causes: Watercore is particularly bad in sections where heat and sunlight are intense. High temperatures at the time the apples are approaching maturity favor its development. The disease is most likely to occur in fruits that are freely exposed to sun. A severe form of the disease, visible watercore, is frequently associated with sunburn.

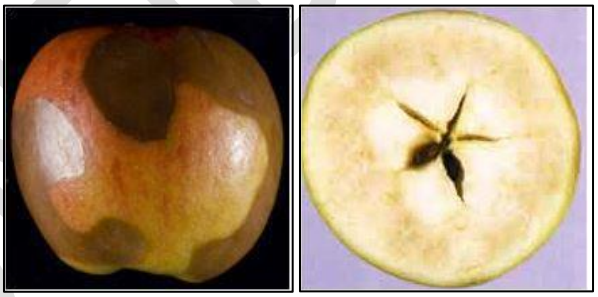


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Core Flush*:* It is characterized by turning the flesh into a brown color with a corky texture.



Chilling / Freezing Damage: Fruit stored at temperatures below their critical point may show a variety of symptoms depending on the variety, temperature and duration of storage. Symptoms may vary from browning of the skin, flesh browning and sometimes translucency. In general, chilling injury may occur at temperature below 0°C.



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Internal Breakdown: Observed when the fruit has a vitreous and/or brownish appearance. Occurrence is more prevalent when the fruit is fully mature or overripe and it indicates deterioration of the fruit.

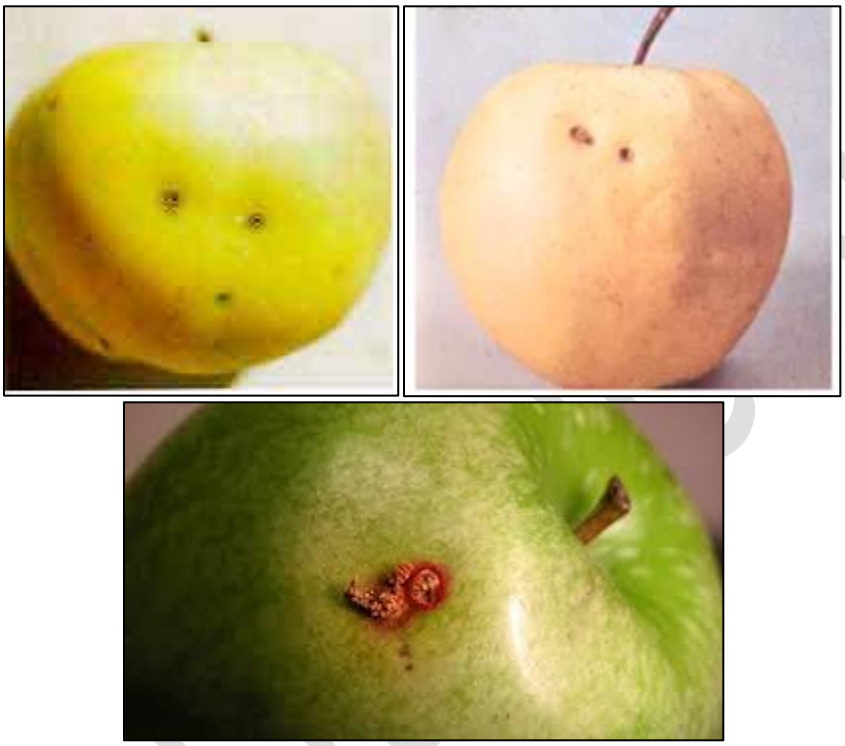


Open Wound*:* Perceived as cuts or punctures in the fruit. Mostly due to harvest or post-harvest handling of the fruit.



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Insect Presence / Damage: It is determined by affected skin showing tiny black or brown blotches on the peal of the fruit and the fruit flesh, reducing shelf life.



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